

- 6 Fig. 6.1 is a simplified diagram representing a section through the human immunodeficiency virus (HIV) particle that causes HIV/AIDS. The diagram shows the virus particle about to attach to the cell surface membrane of a T-helper cell at a receptor protein called CD4. A second protein (coreceptor) called CCR5 is also necessary for the virus particle to enter and then infect the T-helper cell.

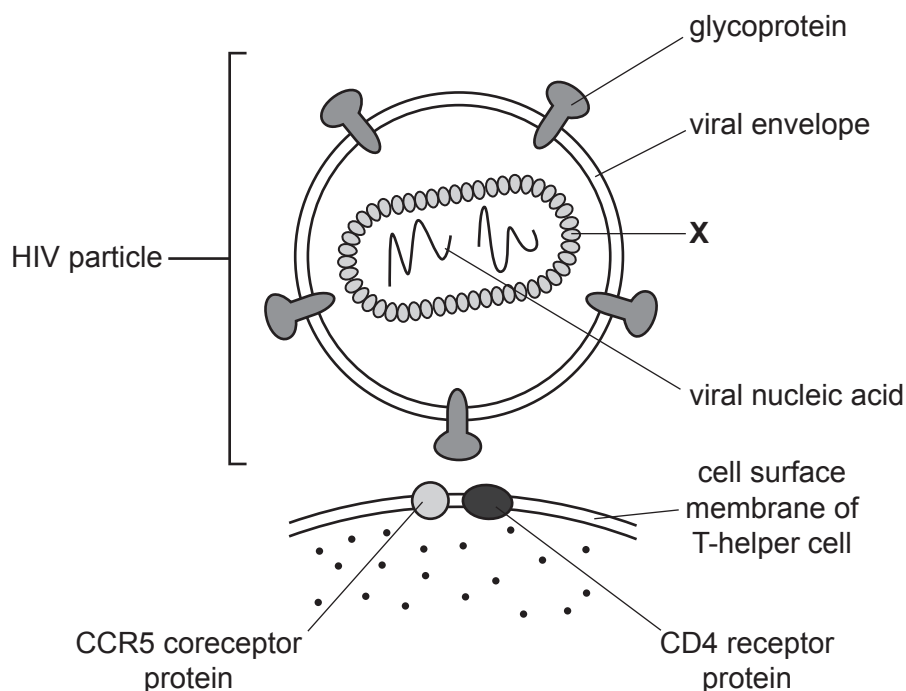


Fig. 6.1

- (a) Identify structure **X** in Fig. 6.1.

..... [1]

- (b) Explain how the ability of the immune system to resist the damaging effects of a pathogen is affected by destruction of T-helper cells.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (c) Studies have shown that some individuals did **not** become infected with HIV even though they were repeatedly exposed to the virus. Later discoveries indicated that these individuals had a mutation in the gene for the CCR5 coreceptor protein.

Suggest how mutation of the gene for the CCR5 coreceptor protein provided protection against HIV infection.

.....

.....

..... [1]

- (d) The use of monoclonal antibodies against the CCR5 coreceptor protein (anti-CCR5) has been shown to be effective in the treatment of HIV infection.

Outline how anti-CCR5 monoclonal antibodies can be synthesised in the laboratory using the hybridoma method.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 8]